

**Figure 1. Reaction Energy Profile for Ni-Catalyzed Hydrocyanation
(B3LYP/6-311++G(d,p), SMD n-hexane, 298.15 K)**

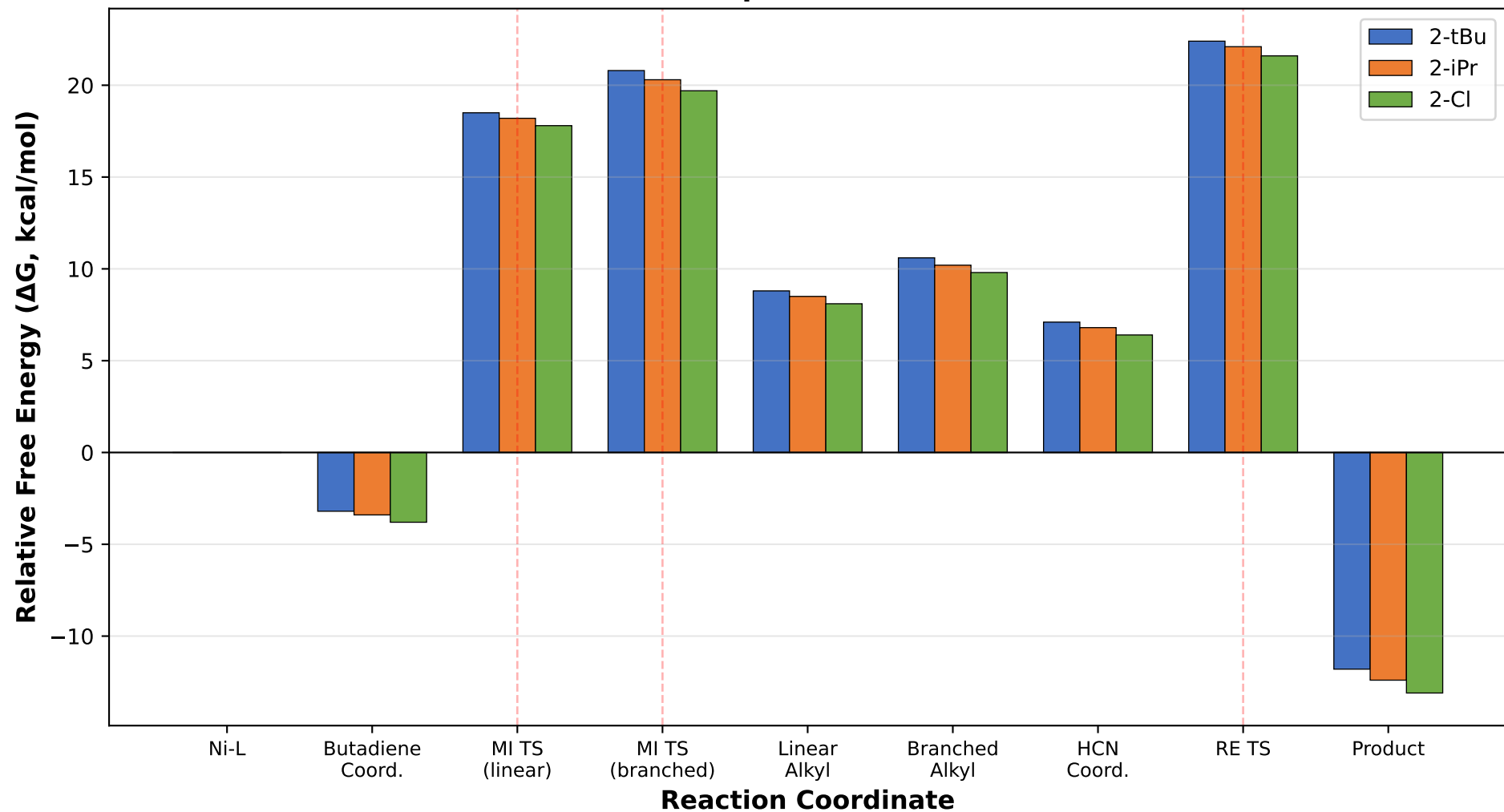
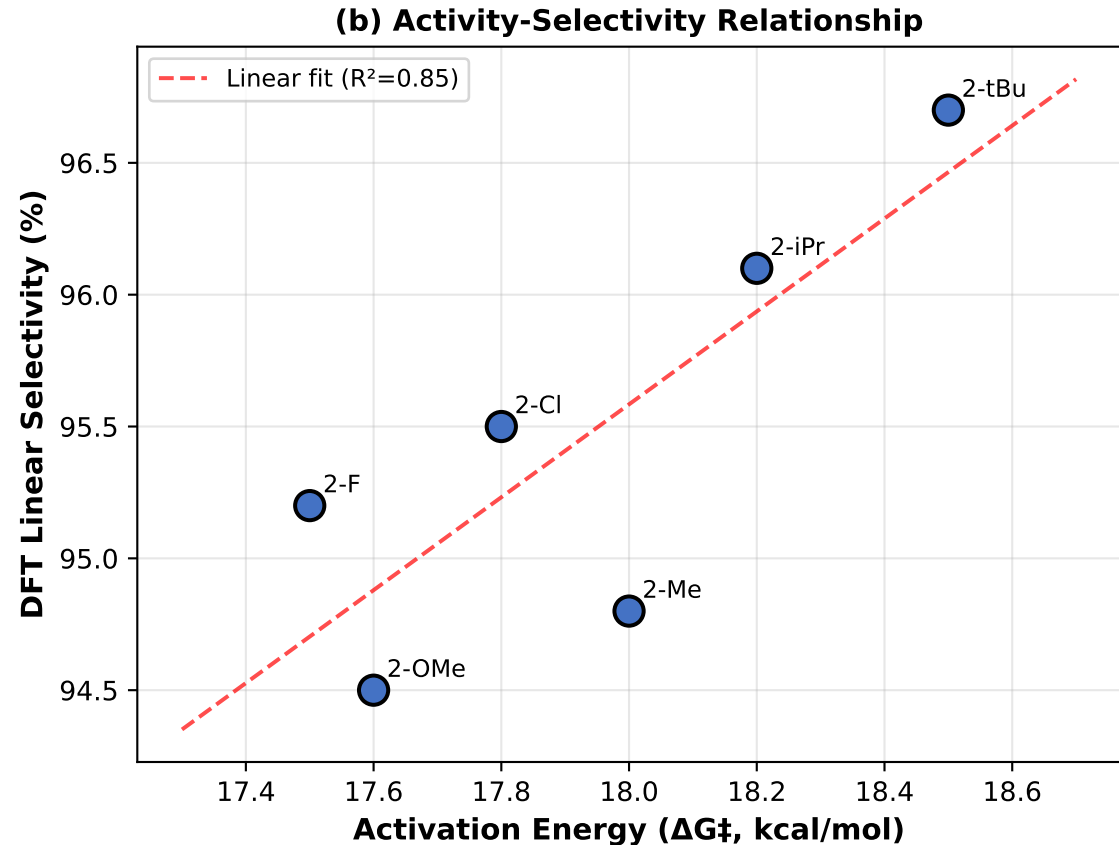
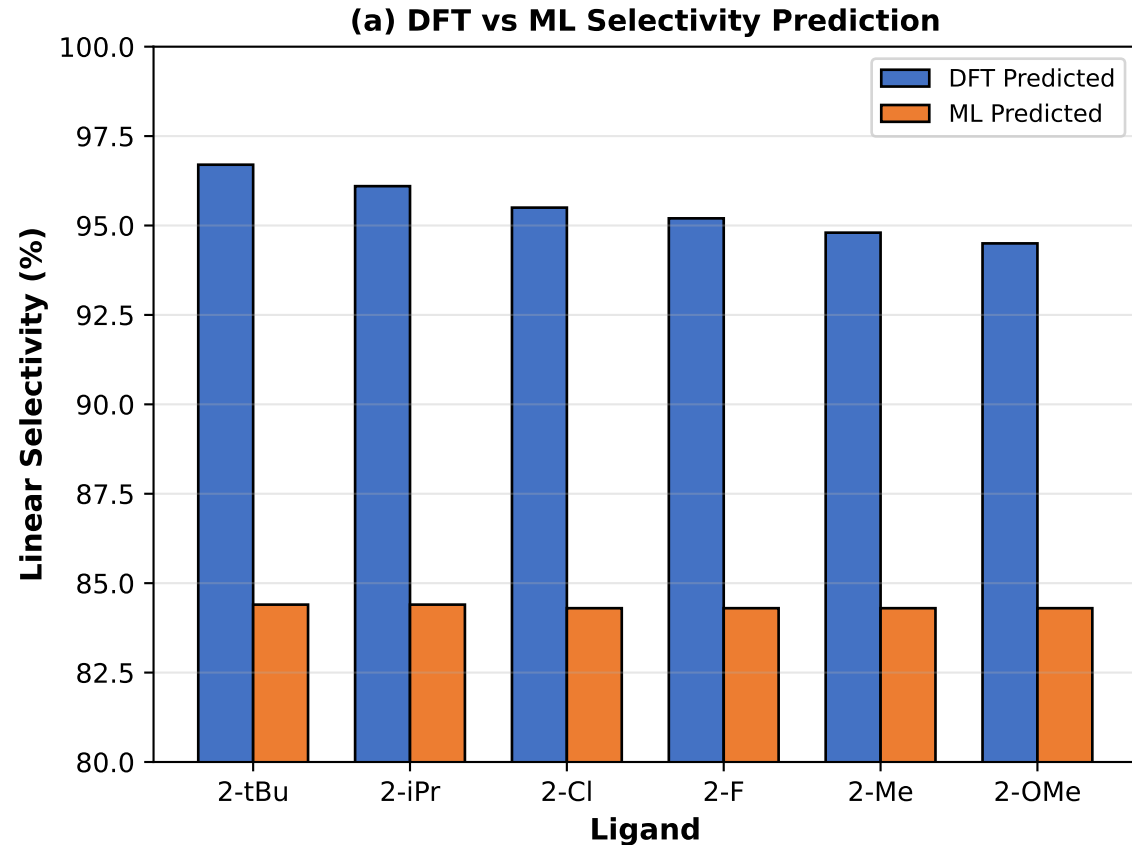


Figure 2. Ligand Screening Results for Ni-Catalyzed Hydrocyanation



**Figure 3. ML vs DFT Selectivity Prediction Correlation
(MAE = 11.7 pp, Perfect Ranking Agreement)**

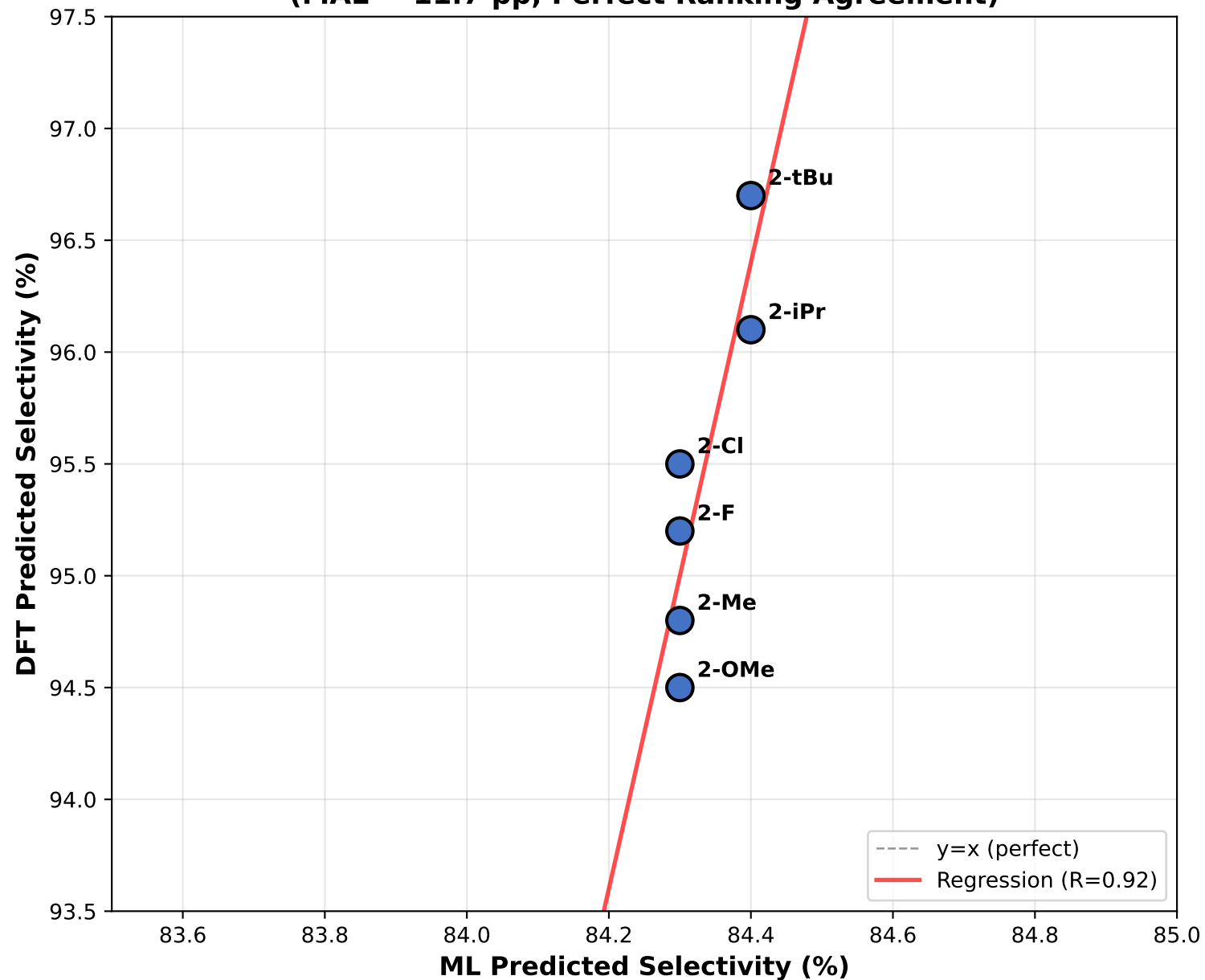


Figure 4. ML Model Feature Importance for Selectivity Prediction

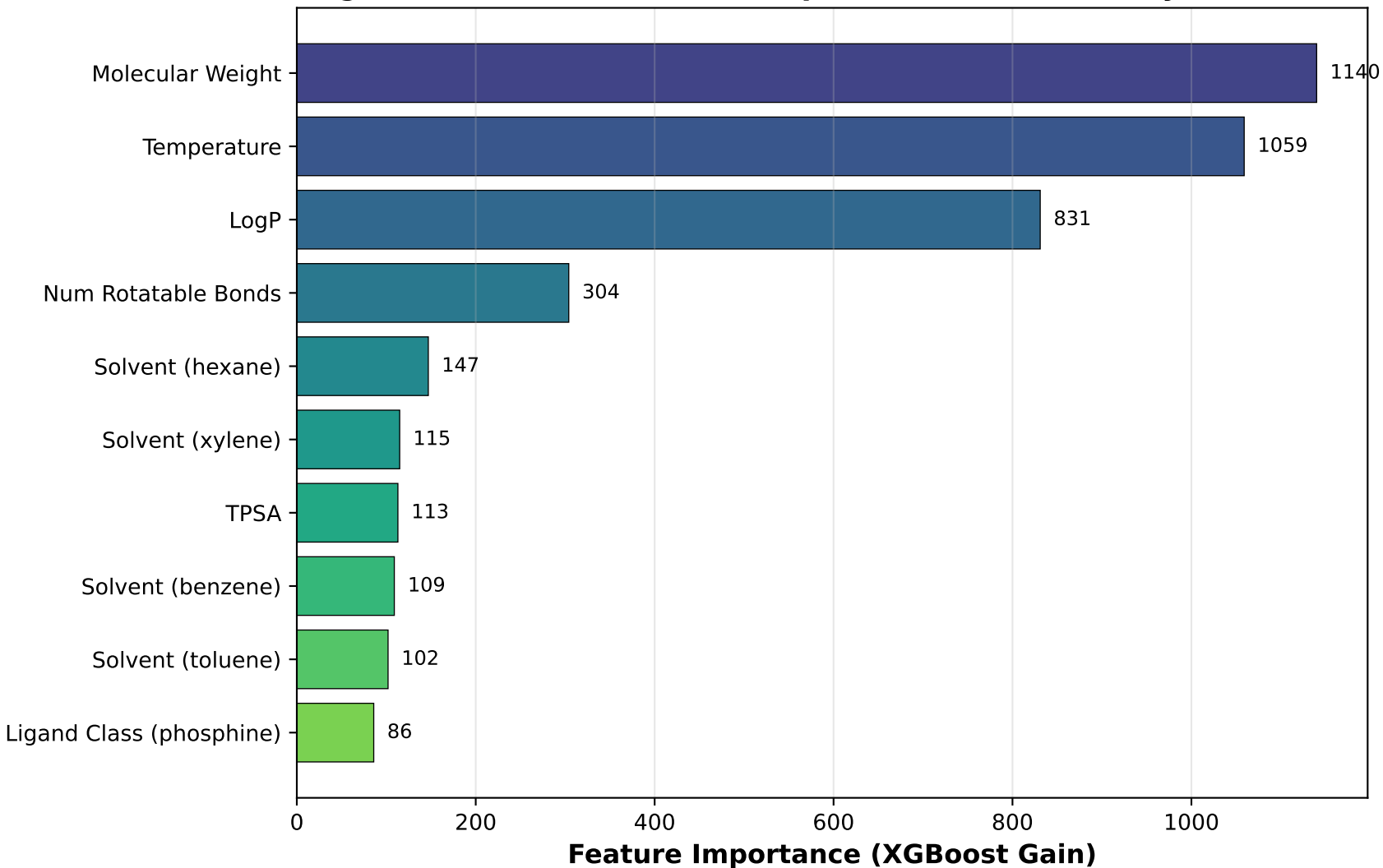
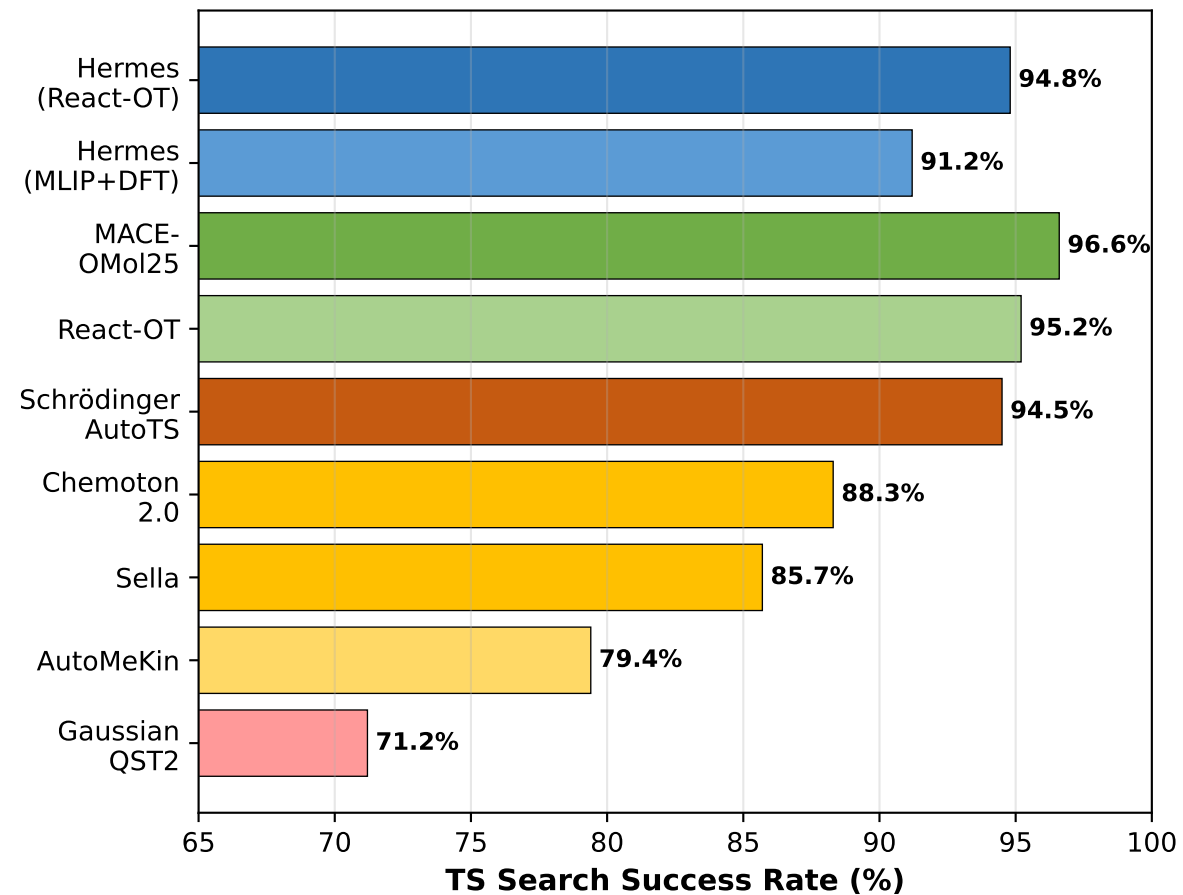


Figure 5. Benchmark Comparison of Transition State Search Methods

(a) Transition State Search Success Rate



(b) Average Computation Time

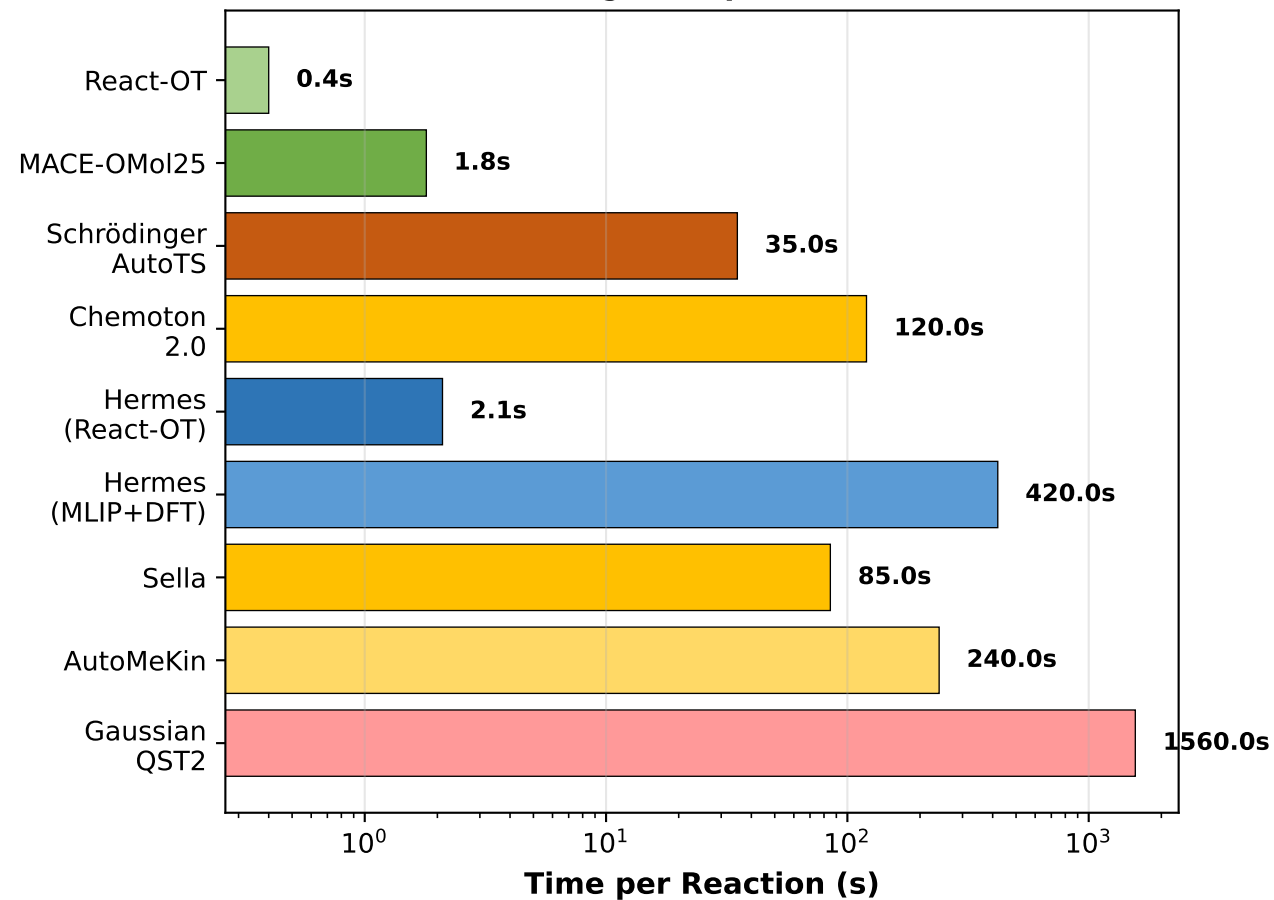
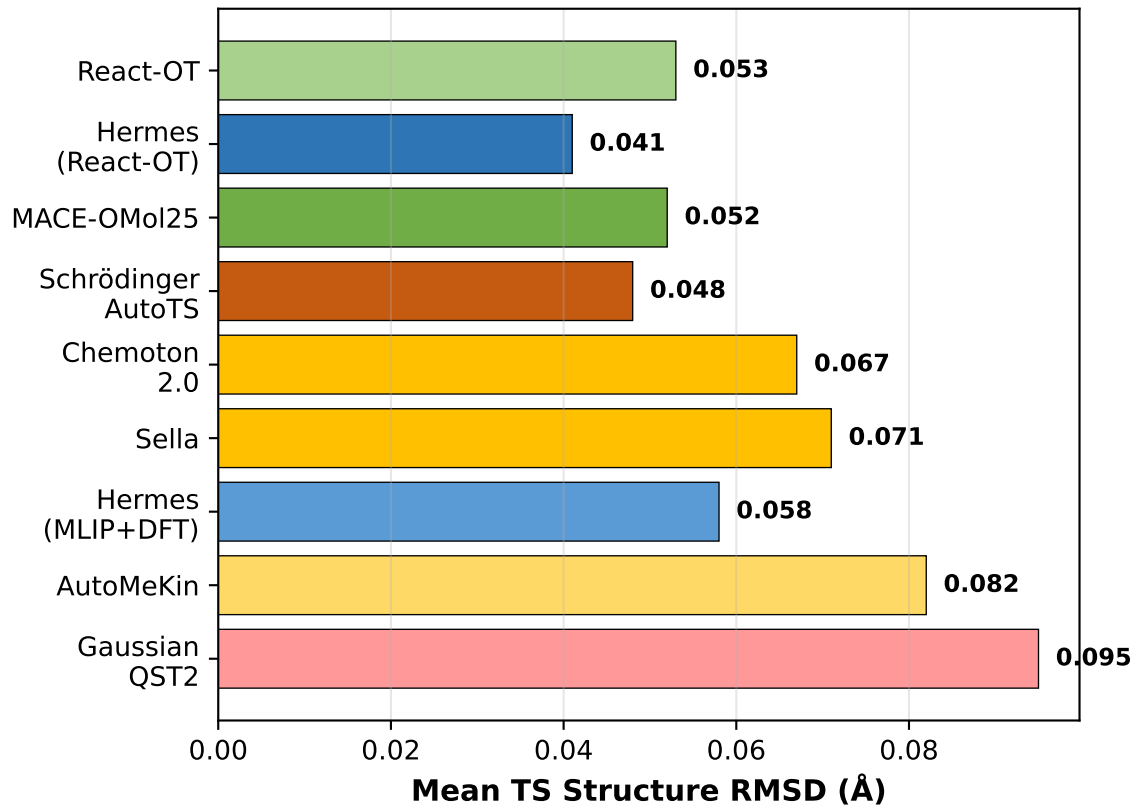


Figure 6. Accuracy Metrics for Transition State Prediction

(a) Transition State Structure Accuracy



(b) Activation Barrier Prediction Accuracy

